

Write your questions
and thoughts here!

Rationalize Fractions with Radicals

1. $\lim_{x \rightarrow 5} \frac{\sqrt{x+4}-3}{x-5}$

2. $\lim_{x \rightarrow 10} \frac{x-10}{3-\sqrt{x-1}}$

Complex Fractions

3. $\lim_{x \rightarrow 0} \frac{x}{\frac{1}{x-4} + \frac{1}{4}}$

4. $\lim_{x \rightarrow 0} \frac{\frac{1}{(x+3)^2} - \frac{1}{9}}{x}$

1.7 Selecting Procedures for Determining Limits

Calculus

Evaluate each limit.

1. $\lim_{x \rightarrow 0} \frac{\sqrt{x+7}-\sqrt{7}}{x}$

2. $\lim_{x \rightarrow 1} \sqrt{x+4}$

3. $\lim_{x \rightarrow 0} \frac{\frac{1}{x+3} - \frac{1}{3}}{x}$

$$4. \lim_{x \rightarrow -3} \frac{x-2}{x^2-3x+2}$$

$$5. \lim_{x \rightarrow 7} \frac{\sqrt{x+9}-4}{x-7}$$

$$6. \lim_{x \rightarrow 0} \frac{2x^5+3x^4}{x^4}$$

$$7. \lim_{x \rightarrow -6} \frac{x}{\frac{1}{x+6}-\frac{1}{6}}$$

$$8. \lim_{x \rightarrow 5} \frac{x^2-5x}{x-5}$$

$$9. \lim_{x \rightarrow 0} \frac{\sqrt{x+9}-3}{x}$$

$$10. \lim_{x \rightarrow 1} \frac{\frac{1}{3}-\frac{1}{3x}}{x-1}$$

$$11. \lim_{x \rightarrow 0} \frac{\sqrt{x+11}-\sqrt{11}}{x}$$

$$12. \lim_{x \rightarrow 3} \frac{\sqrt{2x-6}}{x}$$

$$13. \lim_{x \rightarrow 0} \frac{\frac{1}{(x+2)^2}-\frac{1}{4}}{x}$$

1.7 Selecting Procedures for Determining Limits

Test Prep

$$14. \lim_{x \rightarrow b} \frac{b-x}{\sqrt{x}-\sqrt{b}} \text{ is}$$

(A) $-2\sqrt{b}$

(B) $-\sqrt{b}$

(C) $2b$

(D) $\sqrt{b}$

(E) $2\sqrt{b}$